

Received 14 June 2025, accepted 4 July 2025, date of publication 7 July 2025, date of current version 25 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3586804

RESEARCH ARTICLE

Priority-Aware Multi-Agent Deep Reinforcement Learning for Resource Scheduling in C-V2X Mode 4 Communication

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This work was supported in part by the Project MOST-Sustainable Mobility National Research Center; and in part by the European Union Next-GenerationEU (Piano Nazionale di Ripresa e Resilienza (PNRR)–Missione 4 Componente 2, Investimento 1.4–D.D. 1033 17/06/2022) under Grant CN00000023.

ABSTRACT The need for vehicular networks with exceptional levels of reliability and negligible delay in communication, especially with the ongoing 5G and the upcoming generation of 6G systems, has given rise to Cellular-Vehicle-to-Anything C-V2X systems. This paper proposes a novel framework of Priority-Aware Multi-Agent Deep Reinforcement Learning (PA-MADRL) to address with issues such as high interference, dynamic allocation of critical signals, resource contention in density traffic, dynamic fading, and safety-critical message ratio. This approach is aimed at the development of an effective and adaptive resource allocation mechanism that will improve the likelihood of transmission of safety critical signals and improve overall network performance in metropolitan areas. PA-MADRL allows autonomous vehicles to efficiently allocate resources despite varied congestion along with interference conditions by combining centralized training for global optimization and decentralized execution for scalable decision-making. PA-MADRL is derived from the concept of optimal global allocation of resources in central training and intelligent instantaneous distribution in decentralized setting. Several tests carried out at different levels of interference and traffic densities demonstrate that PA-MADRL improves throughput by 35%, lowers latency by 40% and increases Packet Delivery Ratio (PDR) by nearly 50%, especially when the traffic density increases. These results show that PA-MADRL has the capability to adaptively work in dense vehicular networks with low communication overhead. Ultimately, the framework enhances safety in autonomous transportation by facilitating the timely delivery of critical messages, thus contributing to sustainable and intelligent transportation systems as part of next-generation smart cities.

INDEX TERMS VANETs, C-V2X, PA-MADRL, intelligent transportation systems, autonomous driving.

I. INTRODUCTION

Recent advancements in intelligent transportation systems (ITS) the integration of automotive sector with that of

automation and connectivity. In this evolution Cellular-Vehicle-to-Anything (C-V2X) communication plays a fundamental role, by allowing each equipped vehicle to communicate with any other equipped entity in real-time, allowing advanced new services such as cooperative and remote driving or timely traffic management systems

The associate editor coordinating the review of this manuscript and approving it for publication was Abderrezak Rachedi¹.

necessary for the genesis of autonomous vehicles and smart cities [1]. By virtue of the integration with 5G and the predicted future research of 6G, C-V2X systems will be able to operate under ultra reliable low latency communication which serve the need of the next generation vehicular communication networks [2]. These advancements allow unprecedented levels of connectivity and the ability for real time responsiveness in an autonomous vehicle [3]. However, the traditional ways of resource distributions are not efficient enough, yet, to support these services, in contrast to the flexible scheduling or rules grounded on theory. In fact, current methods are typically unable to incorporate real-time traffic and network variations and are further unsuitable especially in heavily utilized networks and conditions [4], [5]. Hence, they lack mechanisms to schedule both safety critical and non-safety critical messages, which results to higher message delays, low message delivery rates and poor system response time when operated under interference. However, existing studies have paid limited attention to incorporate priority-based scheduling functionalities into intelligent and adaptive learning systems, which is a big drawback to maintain a reliable and efficient communication in C-V2X networks. Limitation is more imposed in an autonomous car environment, where efficient and accurate delivery of important messages can be a factor of safety [6]. The use of Reinforcement Learning (RL) especially Multi-Agent Deep Reinforcement Learning (MADRL) seems to present a potential solution to these challenges. Since MADRL empowers distributed actors (such as vehicles) to acquire local optimal decision-making mechanisms from the environment in which they exist and evolve by interacting with it, it can smoothly bring modifications to an increasing network occupation. However, at the best of the authors' knowledge, existing research provides limited attention to the extension of Physical Sidelink Access (PSA) enabled lower-layer priority-aware scheduling mechanisms for effectively and reliably providing end-to-end C-V2X network connectivity and timely delivery of safety critical messages [7], [8], [9]. Furthermore, the resource management techniques used in the C-V2X networks are normally either centralized or distributed. In centralized systems a controller deals the overall network performance, but it is usually difficult to achieve scalability and latency on a large scale if there are shifting parameters present. On the other hand, distributed strategies share decisions at the vehicle level, which leads to faster, more scalable solutions though not global optimal. To solve these gaps, this paper proposes the Priority-Aware Multi-Agent Deep Reinforcement Learning (PA-MADRL) framework that incorporates priority-aware scheduling using the combination of employing multi-agent RL for efficient resource distribution in C-V2X Mode 4 communication system. The proposed framework takes advantage of both a centralized for the training process and a decentralized deployment of vehicles, enabling self-organizing optimization of resource over the prioritized delivery of critical messages. In doing so, this work guarantees that while

having the flexibility and expandability needed for modern application, MADRL effectively conveys information, minimizes interference, and optimizes resources in dense city settings. In addition to providing a state-of-the-art solution for the allocation of resources in vehicular networks, this research opens up a path toward formulating optimal and dependable communication architecture for 5G and even for the subsequent anticipated 6G self-driving transportation system. In summary, this paper propose PA-MADRL framework for resource scheduling in C-V2X Mode 4 networks. Our approach introduces real-time, multi-tier message prioritization into the decision-making process of each agent (vehicle), supported by a centralized training with decentralized execution paradigm. This design allows agents to learn coordinated policies using global information during training, while enabling fast, localized decisions at runtime crucial for practical deployment in dynamic vehicular scenarios.

The remainder of this paper is organized as follows: Section II reviews the related literature on resource allocation in C-V2X networks. Section III presents the system model and problem formulation. Section IV details the proposed PA-MADRL framework. Section V discusses simulation results and performance evaluation. Finally, Section VI concludes the paper and outlines future research directions.

II. LITERATURE SURVEY

This section presents outlines a comprehensive analysis of the literature structured into three distinct subsections. The first subsection, labeled A, focuses on Centralized Approaches. The second subsection addresses Decentralized Approaches, while the third subsection explores Hybrid Approaches. Recent advancements in resource allocation for Cellular Vehicle-to-Everything (C-V2X) networks have been driven by state-of-the-art methodologies such as Deep Reinforcement Learning (DRL), Multi-Agent Systems (MAS), and priority-based frameworks. These approaches are typically categorized based on network architecture into centralized, decentralized, and hybrid frameworks, depending on the resource allocation strategy and network requirements. However, many of these studies fail to address the challenges posed by highly dynamic and dense vehicular environments, where traffic patterns and network conditions change rapidly. Below, we critically analyze the existing literature, highlighting these gaps and justifying the need for the contributions of this study.

A. CENTRALIZED

An important investigation of centralized solution for resources allocation in C-V2X systems has been done in [10] which examines centralized resource allocation utilizing spectrum re-partitioning particularly in highway settings, is a noteworthy contribution to centralized resources allocation for C-V2X networks. This approach dynamically reconfigures the spectrum resources based on real-time traffic conditions to enhance communication performance and reduce interference. By centralizing the decision-making

process, the system can optimize network performance across a wide area, particularly in high-density environments such as highways. The results demonstrate improved throughput and communication reliability in fluctuating traffic conditions, although the scalability of such a solution remains a challenge in highly dynamic and large-scale networks. On the other hand, in [11], the authors propose a low-complexity centralized deployment of resources for dense C-V2X networking. They address the resource allocation problem by formulating it as a mixed-integer non-linear non-convex refinement challenge, which is challenging to solve optimally in large-scale networks. To overcome this, the authors introduce sub-optimal matching algorithms that reduce computational complexity while still delivering efficient resource allocation. This strategy is particularly Constructive in dense C-V2X environments, where high numbers of vehicles may cause congestion and interfere with communication reliability. The results show that their introduced method Markedly advances the system's sum-rate performance, achieving near-optimal results while ensuring scalability. This work emphasizes The functionality of centralized resource allocation techniques in enhancing the overall efficiency and performance of automotive networks, especially in high- concentration environments.

The authors in [12] examine the scalability issues of C-V2X communications for both low and high traffic densities. In particular, the authors mention the impact of the 3GPP Release 14 Long Term Evolution C-V2X (LTE-V2X), implemented on the PC5 sidelink interface.. Particularly, when addressing safety applications, the scalability problems involve issues related to network saturation, packet loss, and the ability to ensure robust communication in high traffic vehicle environments. To tackle these problems, the authors take various measures to avail LTE-V2X for commercial deployment by making the communication more reliable and less congested. Moreover, the paper discusses the improvements made on the 3GPP Release 16 that includes the 5G New Radio V2X (NR-V2X) with focus on NR sidelink interface. This makes it possible for V2X to operate more effectively, especially under the anticipated harsh specifications for low latency and high reliability for connected and autonomous vehicle systems in the future.

Another notable study focuses on the use of Stackelberg Game Theory for resource allocation in C-V2X networks, specifically within the Internet of Vehicles (IoV) framework. In this work, [13] propose a novel resource allocation approach that employs a hierarchical decision-making process, where the centralized controller (leader) optimizes resource allocation, whereas the vehicles (followers) adjust their actions based on the leader's strategy. The approach shows significant potential for optimizing network resource usage while maintaining communication reliability and minimizing interference in a competitive environment. This centralized method offers strong performance benefits, particularly in scenarios where real-time, Resource optimization allocation is critical to the functioning of the IoV.

In [14], the authors propose a centralized resource allocation method in C-V2X Mode 3, where the evolved NodeB (eNodeB) is responsible for resource scheduling. The method leverages the exchanged information between vehicles and the base station, optimizing resource allocation and adapting to fluctuating traffic conditions and network demands. This approach demonstrated enhanced system performance, especially under low to moderate traffic densities, with reduced latency and interference. However, scalability and computational overhead remain challenging, particularly in high-density environments. In [15], The authors presents a hybrid spectrum along with power allocation strategy for 5G networks that considers QoS for vehicle transmissions. The centralized framework optimizes resource utilization by considering link adaptation and QoS requirements, ensuring efficient spectrum use while minimizing power consumption. The approach significantly improves network throughput and reduces latency, especially under stringent QoS constraints, making it a promising candidate for C-V2X applications in controlled settings.

B. DECENTRALIZED

One recent research works has explored the decentralized Managing resources problem in decentralised platoon-based C-V2X networks using multi-agent reinforcement learning (MARL) to mitigate the (Age of Information) AoI. This method allows for the optimal handling of resources by the vehicles through the use of local information by the agents in which they are employed, and the timely and efficient communication necessary in real world situations. Similar other findings from the study on dynamic multi-agent systems also explain the advantage of decentralised approach in which agents self organise and make decision with their own information. This approach increases the scalability and flexibility in rapidly evolving networks conditions [16]. The work in [17] focuses on the evaluation of how the packet generation interval influences sidelink C-V2X in autonomous mode, with special attention to decentralized communication architecture. The work also tries to assess the cost of having shorter and longer intervals between generating new routes, in relation to delays, loaded and overall network utilization. It notes that shorter intervals are useful for minimizing the time taken to communicate and hence increase the efficiency of the flow of information;The downside is that such a setup often encourages congestion leading to further deterioration of the network in areas where traffic is dense. Conversely, longer intervals alleviate congestion but result in higher latency, potentially compromising the instantaneous requirements of critical vehicular applications. This study highlights the need of adaptive algorithms that have the capability to perform dynamic packet generation interval, which is crucial for ensuring optimal performance in dense and highly dynamic environments, where network demands fluctuate rapidly.

Moreover, the ideas of decentralized resource allocation using the MARL method have been studied in dynamic

environment such as the Internet of Robotic Things (IoRT). This structure, in fact, empowers the agents to operate autonomously since they depend on distributed conditions only, besides eliminating resource redundancy as opposed to the authority commanding at the higher levels. It has been shown that this decentralized architectural approach is indeed viable for achieving goal of scalability, flexibility and real-time response in complex and large scale systems such as IoRT. As a result, MARL enhances cooperation and efficient utilization of resources within devices that are interrelated, and it is applied effectively in extensive and sophisticated numerous-/self-agent ecosystems [18]. On the other hand Recent research has revealed that decentralized systems were more effective, in particular, multi-agent reinforcement learning (MARL) for managing the resources within the realm of vehicular communication networks. These decentralized strategies allow individual vehicle to decide its maneuver according to the local network environment, which can provide large scale and dynamic advantages due to its structure. MARL also decentralizes its structure meaning that in the worst case congestion, decentralized network eliminates network latency which would improve the network's capability in prioritizing alert public notification especially from emergency vehicles. This method also improves resource usage in real-time and is reliable across the network and its components [19].

As an extension, [20] explored adaptive resource reservation interval (RRI) selection algorithms to enhance awareness of cooperation in decentralized New Radio -V2X systems. Their proposed methods enable vehicles to make localized decisions regarding RRI, reducing communication delays and improving tracking accuracy in dense traffic scenarios. The decentralized nature of the approach enhances system scalability and responsiveness under high mobility and dynamic traffic conditions. However, ensuring global resource optimization remains a critical challenge for such decentralized strategies.

Additionally, [21] focused on joint spectrum and power allocation in V2X communications, addressing the challenges posed by imperfect channel state information (CSI). Such framework enables vehicles to make their own decisions and increases the robustness and adaptability of the system in diverse vehicular environment. The results of the investigation showed that the developed technique provides for certain degree of communication reliability and effective utilization of resources in distributing traffic among nodes of the network particular in stable traffic scenarios with rapidly changing conditions. As the case studies indicate, these decentralized methods are being actively implemented in C-V2X systems, mostly through distributed learning approaches. A self-organizing resource allocation in C-V2X networks multi-agent Q-learning algorithm was proposed in [22]. This strategy optimizes subcarrier and power allocation by decoupling Partition the problem into two sub-issues and allowing vehicles to make local decisions. The results show an 18% improvement in energy efficiency

over traditional methods, demonstrating the effectiveness of decentralized learning approaches in dynamic and large-scale environments.

C. HYBRID

C-V2X enhances automakers ability to implement many techniques like (In-Vehicle Information) IVI, entertainment, and social communications which are hard to operate in complex dynamic systems like road networks with heavy traffic. Some of the problems are related to resource management within C-V2X communications networks. Others include the capacity to coordinate actions of multiple systems (or) the ability to integrate global and local decision making as addressed by GMRL in the study done by [23] However, as GMRL shows, balancing global and local strategies is essential for resource management in traffic service C-V2X. That ensures more effective communication within the system.

Moreover, the aspect of resource allocation in the Cellular Vehicle-to-Everything (C-V2X) systems is also well researched and such techniques have been shown to work with reinforcement learning (RL) approaches as well. More specifically, one research [24] aims at the Joint Channel and Power Allocation problem in C-V2X networks, by utilizing several decentralized decision making multi-agent deep reinforcement learning (DRL) strategies. Reinforcement learning with DRL has been coupled with Graph Neural Networks (GNN) and attention layers in order to boost the productivity of resource allocation and minimize the amount of signalling. And, multi-agent federated drl models have been studied for different types of traffic with the goal of achieving faster convergence and improved generalization. So, results presented in this study indicate the efficacy of the DRL based methods in the optimization of resource distribution in vehicular scenarios that are dynamic in nature.

Research on resource allocation in Vehicle Networks (VNs) and Multi-Agent Systems (MAS) has witnessed a rise especially regarding the integration of central and local control systems Such structures make it possible for multi-agent systems, for instance, vehicles in C-V2X networks, to perform neighboring interactions, take local actions, and simultaneously be monitored by a central authority for control purposes. Such hybrid frameworks have been successfully applied to improve resource utilization and network performance in dynamic environments. A notable example is the work presented in [25], where the integration of centralized and decentralized methods in C-V2X networks demonstrated considerable enhancements in both Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication. These findings highlight the potential of hybrid approaches to optimize resource management in complex, high-traffic environments.

Recent research on Spectrum management in Vehicular Networks (VNs) has focused on hybrid Multi-Agent Reinforcement Learning (MARL) approaches, consolidating integrating centralized and decentralized learning techniques.

A notable contribution is made by [26], where the authors proposed a hybrid framework that integrates Deep Q Networks (DQNs) with the QMIX technique to enhance resource allocation in V2X communication systems. In this approach, individual vehicles (V2V agents) make local decisions while collaborating through a centralized QMIX network, which aggregates the actions of all agents. This enables dynamic adaptation to real-time traffic conditions, optimizing spectrum sharing and augmenting overall functionality of the network. The results demonstrate significant gains in V2V signal propagation probability and V2I information exchange, showing the potential of hybrid MARL models for efficient resource management in dense, high-traffic environments.

The critical review of existing literature reveals significant gaps that justify the necessity of our proposed research. While centralized approaches [10], [11], [12], [13], [14], [15] demonstrate effectiveness in controlled environments through spectrum re-partitioning and optimized resource scheduling, they fundamentally lack the flexibility required for dynamic priority-aware resource allocation in real-world vehicular scenarios. These methods treat all communications with equal importance, failing to address the critical requirement for differentiated handling of safety-critical messages versus routine data transmission. In contrast, decentralized approaches [16], [17], [18], [19], [20], [21], [22] offer enhanced scalability and adaptability through multi-agent reinforcement learning and distributed decision-making mechanisms, yet they suffer from the inability to ensure system-wide priority-based optimization, particularly under high-density traffic conditions where global coordination becomes essential for safety-critical applications. Existing hybrid methodologies [23], [24], [25], [26] attempt to bridge this gap by combining centralized oversight with decentralized execution, demonstrating improvements in both V2V and V2I communications. However, these approaches consistently overlook the integration of priority-aware mechanisms that are fundamental for C-V2X networks serving autonomous and connected vehicles. Unlike previous studies that focus primarily on throughput optimization, latency reduction, or energy efficiency as standalone objectives, our research addresses the critical gap by incorporating multi-tier priority classification within a hybrid framework that maintains the scalability benefits of decentralized approaches while ensuring global priority-based resource optimization through centralized coordination.

Our contribution distinguishes itself from existing literature through three fundamental aspects: First, unlike [10], [11], [12], [13], [14], [15], [23], [24], [25], and [26], our approach integrates real-time message priority classification that dynamically adapts resource allocation based on both traffic conditions and message criticality levels. Second, in contrast to [16], [17], [18], [19], [20], [21], and [22], our hybrid framework ensures that decentralized decision-making does not compromise system-wide priority enforcement for safety-critical communications. Third,

advancing beyond all reviewed studies, our methodology provides practical implementation strategies for real-world C-V2X deployments where theoretical optimization must meet the practical requirements of mixed-traffic scenarios involving emergency vehicles, autonomous systems, and routine communications. Addressing these challenges, our framework fills a crucial gap in the literature: the lack of scalable, adaptive, and priority-sensitive resource management mechanisms capable of guaranteeing timely delivery of safety-critical messages in dynamic, high-density vehicular environments.

III. CONTRIBUTIONS

In addition to extending the existing literature in this area, this research provides key advances in the burgeoning field of ITS, with the following main contributions:

- 1- We propose a novel priority based resource control scheme that enforces high priority messages like CAM and DENM while maintaining high channel throughput and low interferer levels.
- 2- We present a groundbreaking semi-centralized scheduling procedure specifically for C-V2X Mode 4 in order to deal with problems that arise in complex dense urban scenarios. Specifically, our approach adapts to the traffic density and can organize interaction between both autonomous and non-autonomous vehicles in intricate urbanized network.
- 3- We implement centralized training and decentralized implementation, making self-driving cars capable of operating on the basis of observations made in real-life (while running on the road), in response to changing traffic conditions, without, however, requiring a direct interaction with a centralized command and control center.
- 4- Comprehensive Evaluation in Realistic Scenarios: We validate our framework through extensive simulations across diverse traffic conditions, including high-density and mixed-traffic scenarios. The results demonstrate consistent improvements in latency, packet delivery ratio (PDR), and channel utilization, confirming the model's effectiveness and suitability for real-world deployment.

IV. METHODOLOGY

This section presents the developed methodology. Specifically, sub-section I introduces the system model and the problem formulation, sub-section II describes the resource allocation mechanism and finally, sub-section III presents the scheduling strategy employed for C-V2X communication.

A. SYSTEM MODEL

As depicted in Figure 1, we consider a C-V2X Mode 4 network in which various vehicular agents engage in V2V and V2I communication to broadcast both safety and non-safety messages. The system leverages PA-MADRL to allocate resources efficiently, aiming to prevent message collisions while prioritizing safety-critical messages. The allocation of communication resources is dynamically adjusted grounded

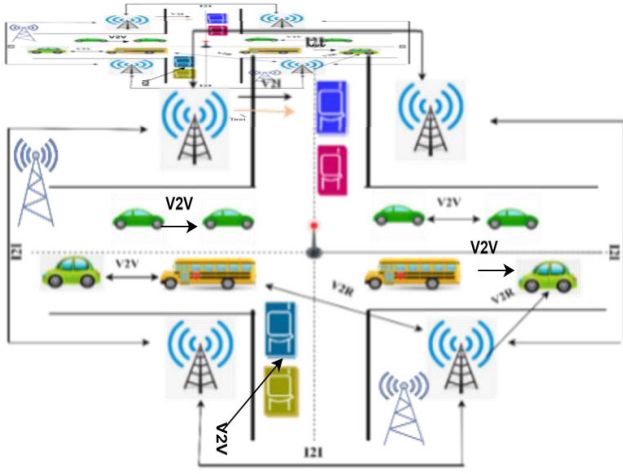


FIGURE 1. System model and framework architecture [27].

in real-time traffic conditions and network scenarios inherent to the system. The model gives particular emphasis on the optimal utilization of communication resources, prioritizing messages based on their criticality and the current state of the vehicles. This system model utilises a combination of centralised training for global optimisation, and decentralised reinforcement learning for agile decision making to provide the optimal use of resources in densely populated regions.

1) COMMUNICATION LINKS

The exchange of information between vehicles or between vehicles and the roadside network happen as described in the following:

V2V communication: the announced Vehicle-infrastructure Cooperation (AViC) and DENM are transmitted by vehicle to vehicle, and the Cooperative Awareness Messages (CAM) and Emergency Vehicle Notification Messages (EVNM) are transmitted by the group of vehicles to each other. These messages serve crucial functions such as collision alerts, traffic information sharing, and supporting safety-critical decision-making processes.

Vehicle-to-Infrastructure (V2I) Communication: C-V2X, Integrating a Cellular-Vehicle to Everything (C-V2X) controller (also infrastructure), can directly transmit sensing and coordinating by best-scheduling the available media resources. Resources are managed by the controller according to vehicle needs and conditions of the network. This communication link ensures efficient resource management and prioritization of safety-critical messages while minimizing interference with non-critical transmissions.

2) KEY PARAMETERS

The system operates with the following key parameters:

Vehicles: Each vehicle $v_n \in \{1, 2, \dots, N\}$ transmits messages with specific priority levels. Vehicles act as agents, making decisions on resource allocation based on their state at each time step. **Communication Resources:** The system

uses $\mathbf{M}$ communication resources, represented as $R = \{r_1, r_2, \dots, r_M\}$, where each resource r_m can represent time slots, frequency channels, or power levels.

Message Priority: Each vehicle transmits messages $M_n = \{m_1, m_2, \dots, m_L\}$, and each message m_l has a priority:

$$\text{Priority}(m_l) \in \{0, 1\}$$

- Priority(m_l) = 1 for safety-critical messages (e.g., CAM or DENM),
- Priority(m_l) = 0 for non-critical messages.

State Representation: The state s_v^t of vehicle v_n at time t includes several components influencing its decision on resource allocation. These include:

$s_v^t = \{\text{Priority}(m_l), \text{Traffic Density}, \text{Interference Level}, \text{Available Resources}, \text{Vehicle Speed}\}$

Message Priority is a static property of the message, influencing the resource allocation decisions but not changing dynamically during the simulation.

- **Traffic Density:** The number of vehicles in the communication range, influencing resource contention.
- **Interference Level:** The interference caused by nearby vehicle transmissions.
- **Available Resources:** The available communication resources (time slots, frequency channels).
- **Vehicle Speed:** The vehicle's speed, which affects interference and available resources.
- **Action Space:** The action a_v^t represents the selection of a communication resource $r_m \in R$ at time t . The set of resources R is defined as:

$$a_v^t = r_m \text{ where } r_m \in R = \{r_1, r_2, \dots, r_M\} \quad (1)$$

Each action a_v^t taken by a vehicle influences its future state, such as available communication resources and the interference level. The action selection process, based on the vehicle's state, ensures that the chosen resources optimize network throughput and minimize interference.

The vehicle selects a resource based on its state and the learned policy.

3) KEY EQUATIONS

The system model is built around several key equations that describe how resources are allocated, how interference is computed, and how message priority influences decisions.

3.1. Message Priority:

Message priority determines if a message is safety-critical or non-critical, guiding the vehicle to prioritize critical messages:

$$\text{Priority}(m_l) = \begin{cases} 1 & \text{if message } m_l \text{ is safety-critical} \\ 0 & \text{if message } m_l \text{ is non-critical.} \end{cases} \quad (2)$$

3.2. Interference Level:

The interference level $\text{Interference}_n(r_m)$ for vehicle v_n on resource r_m depends on the simultaneous transmissions by nearby vehicles. It increases with the proximity of other

vehicles and their transmission power. The interference for resource r_m is given by:

$$I(gi, j) = \sum_{i \neq n} \text{Interference}(v_n, v_i) \cdot \frac{P_i}{d_{ni}^\alpha} \quad (3)$$

where:

- Interference Factor (v_n, v_i) is the interference factor between vehicles v_n and v_i ,
- d_{ni} is the distance between vehicles v_n and v_i
- α is the path loss exponent that models signal attenuation over distance.

3.3. Traffic Density:

Traffic density ρ_t is calculated based on the quantity of vehicles within the communication area of vehicle v_n . It is an important factor affecting resource contention and interference:

$$\rho_t = \frac{\text{Number of Vehicles in Range}}{\text{Area of Interest}} \quad (4)$$

where the Area of Interest is the region within which vehicles communicate. As traffic density increases, the contention for communication resources rises, leading to increased interference and potentially reduced system throughput.

3.4. Resource Allocation:

The resource allocation $R_{\text{util}}(v_n)$ for vehicle v_n represents the efficient use of communication resources. It is referred to as the rate of allocated resources to the total available resources:

$$R_{\text{util}}(v_n) = \frac{R_{\text{allocated}}(v_n)}{R_{\text{available}}(v_n)} \quad (5)$$

where:

- $R_{\text{allocated}}(v_n)$ is the resources allocated to vehicle v_n ,
- $R_{\text{available}}(v_n)$ is the total available resources at time t .

B. PROBLEM FORMULATION

The goal of the proposed PA-MADRL framework is to optimize resource allocation in C-V2X Mode 4 networks, where vehicles must communicate efficiently while minimizing interference and latency. The problem is framed as a MADRL problem, where each individual vehicle (agent) makes decisions about resource allocation (e.g., choosing time slots or frequency channels) to achieve the highest possible network throughput, minimize interference, and reduce message transmission latency.

1) OBJECTIVE FUNCTION

The primary goal is to achieve the highest total throughput (reward for all vehicles in the system), while minimizing interference and latency, and ensuring effective use of communication resources. The objective function is constructed as:

$$\max \sum_{i=1}^N \left(\text{Throughput}_i - \lambda \cdot \text{Interference}_i - \mu \cdot \text{Latency}_i + \gamma \cdot \text{Resource Utilization}_i \right) \quad (6)$$

where:

- Throughput i : The successful transmission rate of vehicle i .
- Interference i : The interference caused by vehicle i 's transmission.
- Latency i : The time delay in transmitting messages for vehicle i .
- Resource Utilization i : The extent to which vehicle established the communication resources which are deemed appropriate by percentage i .
- λ , μ , and γ : Weight factors for the respective metrics (interference, latency, and resource utilization), which control the comparative significance of all respective factor in terms of optimization process.

The reward R_t^n for vehicle v_n at time t is defined as:

$$R_t^n = \sum_{m_l \in M_n} \text{Priority}(m_l) \cdot \text{Throughput}_n(m_l, r_m) - \lambda \cdot \text{Interference}_n(r_m) \quad (7)$$

where:

- Throughput $\text{Throughput}_n(m_l, r_m)$ represents the successful transmission rate of message m_l on resource r_m .
- Interference $\text{Interference}_n(r_m)$ is the interference caused by simultaneous transmissions on the same resource.
- λ is a penalty factor to discourage interference.

The goal of the system is to identify a policy which boosts only the throughput of the system and at the same time reduces interference and latency while utilizing all resources available. These weights λ , μ , and γ serve as the controllers of the trade-off between these objectives of the system. These weights can be obtained analytically from simulation results.

2) OPTIMIZATION PROBLEM

The objective is to enhance the overall reward across all the vehicles in the network:

$$\max_{R_n} \sum_{n \in N} \sum_{m_l \in M_n} \text{Priority}(m_l) \cdot \text{Throughput}_n(m_l, r_m) - \lambda \cdot \sum_{n \in N} \sum_{r_m \in R} \text{Interference}_n(r_m) \quad (8)$$

Subject to the following constraints:

$$\sum_{n \in N} \sum_{m_l \in M_n} \text{ResourceUsage}(m_l, r_m) \leq 1, \quad \forall r_m \in R \quad (9)$$

2.1 Interference Minimization:

Every vehicle's transmission interference cannot exceed a certain level so that a network integrity is sustained without compromising the others' signals.

$$\text{Interference}_i \leq \text{Threshold} \quad \forall i \quad (10)$$

2.2 Priority Constraint:

Safety-critical messages resources ought to be assigned initiall:

$$\text{Priority}(m_l) = \begin{cases} 1, & \text{if } m_l \text{ is safety-critical (CAM/DENM)} \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

2.3 Latency Constraint:

The delay in message transmission through each vehicle not exceed beyond the maximum allowable latency especially for safety-critical messages:

$$\text{Latency}_i \leq \text{Max Latency} \quad \forall i \quad (12)$$

3.4 Resource Utilization:

It is essential to make the best use of all available resources while ensuring that bottlenecks do not inhibit performance:

$$\max \sum_{v \in V} U_v \quad (13)$$

This is to avoid the issue where communication resources, For instance, time slots or frequency bands, get allocated inefficiently. In the optimization process though avoid congestion / high latency obtain, throughput and in-interference improved and over-expectation is avoid.

C. FRAMEWORK FOR PRIORITY-AWARE MULTI-AGENT DEEP REINFORCEMENT LEARNING

To address the above challenges, we propose the PA-MADRL framework with the aim to promote an improved perception of the transportation environment in C-V2X Mode 4. This framework encompasses interference control, low latency, and smooth communication among the nodes, which are the key issues for addressing the complications involved in efficient communication in highly developed and densely populated sub urban automobile networks. The integration of priority-aware scheduling ensures that safety-critical messages (for instance, CAMs and DENMs) are transmitted with higher priority than non-safety crucial messages.

The structure used is the Reinforcement learning with deep learning for multiple agents (MADRL) that permits vehicles to make decisions independently of the other vehicles. Additionally, This approach of centralized/decentralized structure bears the benefits of centralized training with decentralized execution features that are effective for real-time and macro execution. This new approach offers a solution that is both efficient and durable for the needs of urban mobile communication, and represents a potential for increasing the safety of autonomous car systems as illustrated in Figure. 2, the PA-MADRL framework consists of two main phases: centralized training and decentralized execution, designed to optimize resource allocation in C-V2X Mode 4 communication systems.

D. SYSTEM ARCHITECTURE

The following essential elements make up the suggested PA-MADRL framework, as illustrated in Figure 3. Priority-Aware Resource Allocation: Ensures that safety-critical

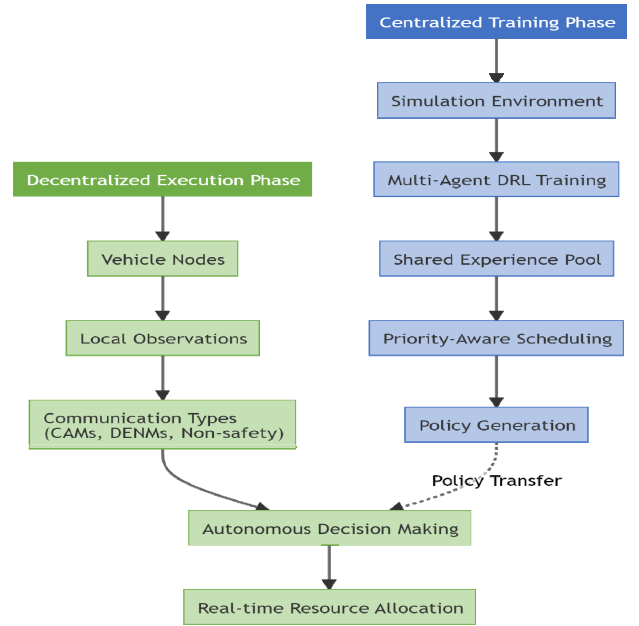


FIGURE 2. PA-MADRL framework.

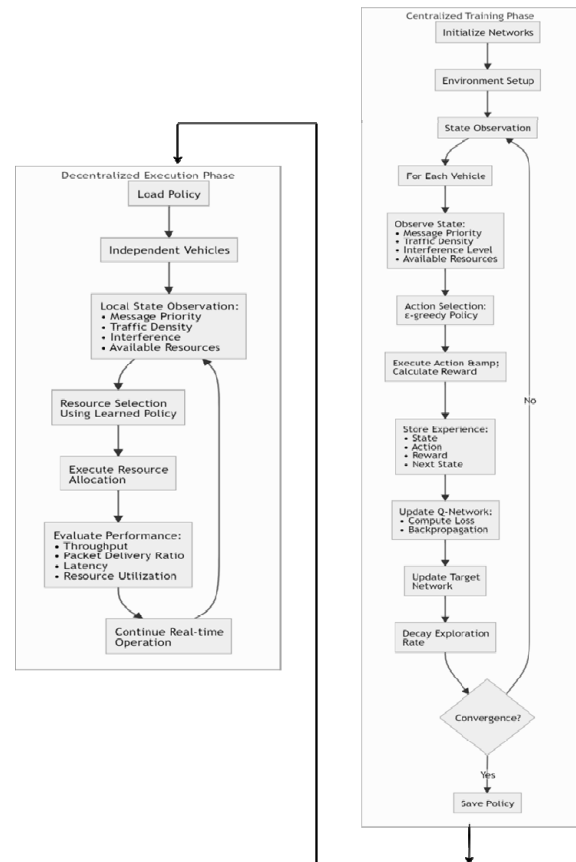


FIGURE 3. System architecture.

messages (e.g., CAMs and DENMs) are prioritized in resource allocation, while non-critical messages are allocated

resources based on network conditions. Multi-Agent Deep Reinforcement Learning (MADRL): Frames all vehicle as a distinct agent making decisions in a shared environment. The agents learn optimal communication policies by interacting with the environment and maximizing cumulative rewards. Centralized Training and Decentralized Execution: Training is performed in a centralized manner using aggregated data from multiple vehicles, while the execution phase allows vehicles to operate independently in real-world scenarios. Fig.3 illustrates the system architecture of the PA-MADRL framework, showing the interaction between priority-aware resource allocation, MADRL components, and the centralized training and decentralized execution phases.

1) FORMALIZATION OF THE APPROACH

The following mathematical models and equations describe the interactions between the various components

1.1. Resource Allocation:

The reward for resource allocation reflects how well the system optimizes both vehicle priority (e.g., latency-sensitive applications) and overall network throughput, guiding the allocation toward efficient and fair use of communication resources. The reward for resource allocation at vehicle v_n at time t is expressed as:

$$R_{t,n} = \sum_{m_l \in M_n} \text{Priority}(m_l) \cdot \text{Throughput}_n(m_l, r_m) - \lambda \cdot \text{Interference}_n(r_m) + \alpha \cdot \text{PDR}_n \quad (14)$$

where:

- $R_{t,n}$: Total reward for vehicle v_n at time t ,
- $\text{Priority}(m_l)$: Priority level of message m_l (1 for safety-critical, 0 otherwise),
- $\text{Throughput}_n(m_l, r_m)$: Throughput of message m_l on resource r_m
- λ : Penalty coefficient for interference,
- $\text{Interference}_n(r_m)$: Interference caused by simultaneous transmissions,
- α : Weight for Packet Delivery Ratio (PDR),
- PDR_n : Packet Delivery Ratio for vehicle v_n , calculated as:

$$\text{PDR}_n = \frac{\text{Number of successful deliveries}}{\text{Total number of packets sent}} \times 100 \quad (15)$$

1.2. State Transition

The environment state transitions as vehicles interact with the communication network, The environment undergoes state transitions in response to vehicle interactions with the communication network, wherein information exchange between vehicles and infrastructure nodes alters the system dynamics The transition function is:

$$s_{t+1}^n = f(s_t^n, a_t^n) \quad (16)$$

where:

- s_{t+1}^n : The state of vehicle v_n at the next time step,
- s_t^n : The state of vehicle v_n at time t ,
- a_t^n : The action taken by vehicle v_n at time t ,

- $f(\cdot)$: State transition function incorporating vehicle dynamics and communication effects.

1.3. Exploration vs. Exploitation

The ϵ -greedy policy balances exploration and exploitation during action selection can be expressed as:

$$a_t^n = \begin{cases} \text{random action} & \text{with probability } \epsilon \\ \arg \max_a Q(s_t^n, a | \theta_v) & \text{otherwise} \end{cases} \quad (17)$$

where:

- a_t^n : Action selected by vehicle v_n at time t ,
- ϵ : Exploration probability.

1.4. Learning Rate Update

The learning rate decays over time to stabilize the learning process is given by

$$\alpha_t = \frac{\alpha_0}{1 + \beta \cdot t} \quad (18)$$

where:

- α_t : Learning rate at time t ,
- α_0 : Initial learning rate,
- β : Decay parameter.

2) ALGORITHM: PRIORITY-AWARE MADRL-BASED RESOURCE ALLOCATION

The PA-MADRL algorithm comprises two stages: Centralized Training and Decentralized Execution Each phase is outlined below with the updated system model and problem formulation. Algorithm 1: PA-DDQN Centralized Training Phase. The Centralized Training Phase of the Priority-Aware Multi-Agent Deep Q-Learning (PA-DDQN) algorithm is responsible for training the vehicles (agents) in a shared environment. During this phase, communication resources are assigned to all vehicles optimally through environmental interaction.

Algorithm 2: PA-DDQN Decentralized Execution Phase

After the Centralized Training Phase, the vehicles operate in a Decentralized Execution Phase, where each vehicle independently allocates resources based on its learned policy. In this phase, vehicles no longer require centralized control and use their learned policies to make real-time decisions regarding resource allocation.

V. SIMULATION RESULTS AND ANALYSIS

A. SIMULATION SETUP

MATLAB and SUMO (Simulation of Urban MObility) have been respectively adopted for the evaluation of vehicular communication systems performance and the modelling of a dynamic urban traffic environment. The simulation is based on a road network that includes urban streets with varying lane configurations. Vehicle movement is modeled using the Random Waypoint Model, with traffic densities ranging from 50 to 500 vehicles/km² and speeds between 30 and 120 km/h. The communication system

Algorithm 1 PA-DDQN Centralized Training Phase**Input:**

- Vehicles $V = \{v1, v2, \dots, vN\}$
- Communication resources $R = \{r1, r2, \dots, rM\}$
- Traffic density and interference levels
- Initial Q-function or policy network θ_v for each vehicle

Output:

- Learned policy network θ_v for communication resource allocation

1. Initialization

- For each vehicle v_n :
- Initialize Q-function ($s_t^n, a_t^n | \theta_v$) and policy network π_v
- Set learning parameters: α (learning rate), γ (discount factor), ε (exploration rate)
- Initialize positive and negative buffers BB and CB

2. Training Loop

- For each episode:
- Reset the environment and states
- For each time step t : Observe state s_t^n and Select action a_t^n
Use ε -greedy policy
- Execute action: Observe next states and reward
- Store experience in appropriate buffer
- positive if $r_t^n > 0$, Negative if $r_t^n \leq 0$.

3. Q-Function Update:

For each experience from buffers: Compute target using Bellman equation:

$$y_v^t = r_t^n + \gamma \max_a Q(s_{t+1}^n, a' | \theta_{-v})$$

- Update Q-function via gradient descent

4. Target Network Update:

- periodically copy weights from θ_v to target network θ_{-v} .

5. Decay exploration rate:

- Gradually reduce ε for less exploration as episodes progress

6. Repeat

- Repeat for a specified number of episodes

7. End

follows C-V2X protocols, operating in the 5.9 GHz frequency band with a 10 MHz channel capacity, supporting the transmission of both safety-critical and non-critical messages. The wireless channel model incorporates Rayleigh fading, path loss, and interference, which are influenced by vehicle density and transmission power. The simulation is executed on a high-performance computational system, equipped with a multi-core processor (Intel Core i7), 32GB of RAM, and an NVIDIA RTX 3080 GPU, enabling efficient execution of deep reinforcement learning tasks.

The system operates on a Linux-based environment, ensuring seamless integration of MATLAB and SUMO. The setup is designed to scale with varying traffic densities, adjusting vehicle counts and network parameters to assess the system's performance under different conditions. This configuration allows for comprehensive analysis of the system's behavior across diverse operational scenarios.

Algorithm 2 PA-DDQN Decentralized Execution Phase**Input:**

- The learned policy network θ_v for each vehicle

Output:

- Autonomous decision-making for each vehicle based on its local state.

Steps:**1. Initialization:**

- Each vehicle loads its learned policy network, which has been trained during the centralized phase.

2. Decentralized Execution:

- For each vehicle v_n in V :
- The vehicle observes its local state s_t^n , which includes
- Message priority, traffic density, interference level, available resources, and vehicle speed.
- The vehicle selects an action a_t^n based on the learned policy or Q-function
- The vehicle transmits its message using the selected communication resource, corresponding to the action.

4. Performance Evaluation:

- After the transmission, evaluate the vehicle's based on:
- Throughput: The rate of successful message transmissions.
- Packet Delivery Ratio (PDR): The ratio of successfully delivered packets to the total number of packets sent.
- Interference: The level of interference caused by simultaneous transmissions on the same resource.
- Latency: The Latency for a message to be transmitted from original to destination

4. Repeat

- Repeat for each time step t until the simulation

5. Ends.**TABLE 1.** Simulation parameters.

Parameter	Value/Unit
Traffic Density	50 to 500(vehicles/km ²)
Vehicle Speed	30–120 km/h
Number of Vehicles (N)	50–200 vehicles
Message Types	Safety-critical (CAMs, DENMs) and non-critical messages
Simulation Duration	1000 seconds
Packet Size	300 bytes
Channel capacity	10 MHz
Carrier Frequency	5.9 GHz
Transmission Power	23 dBm
SINR Threshold	10 dB
Mobility Model	SUMO, Random Waypoint Model
Action Space	Discrete set of resource allocation decisions
Learning Rate	0.01
Discount Factor	0.9

B. SIMULATION RESULTS

This Section presents the performance evaluation of PA-MADRL framework in resource allocation within C-V2X communication networks, compared with known methods

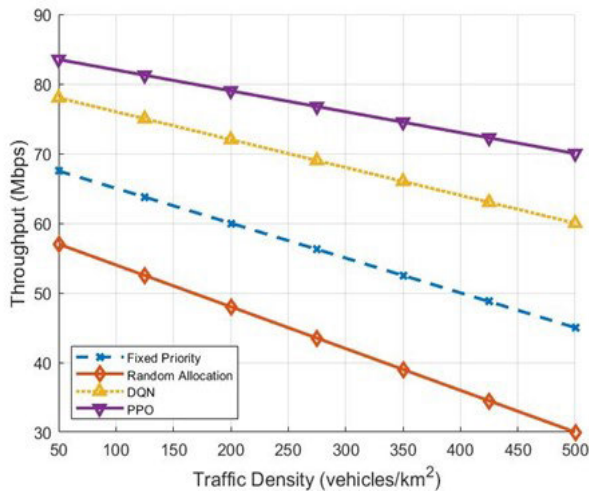


FIGURE 4. Throughput for different schemes.

such as Priority Normalization and Normal Allocation. Simulations were conducted in a simulated urban environment with vehicle densities of 200 vehicles/km² and 500 vehicles/km², representative of typical traffic scenarios in smart cities. PA-MADRL demonstrated strong performance across all evaluated parameters, including throughput, latency, packet delivery rate (PDR), and resource utilization. Notably, the framework improved upon the basic methods by maintaining high performance in densely populated environments, ensuring low latency and high PDR during traffic congestion, and optimizing resource allocation.

Specifically, the centralized PA-MADRL approach achieved superior performance but required additional resources as the number of vehicles increased. Alternatively, the distributed PA-MADRL approach exhibited reduced costs, making it suitable for deployment in large, densely populated cities. The results demonstrate that the PA-MADRL system is capable of effectively managing communication resources in urban environments, ensuring reliable and efficient operations that are critical for smart city safety issues.

In Figure 4, it can be observed that for traffic of 200/km², PA-MADRL reaches 90 Mbps, whereas DQN and PPO reach 65 Mbps and 70 Mbps, respectively. Priority control and random distribution performed well, with output values of 55 Mbps and 50 Mbps respectively. For traffic of 500/km², PA-MADRL maintains 80 Mbps, while the performance of other methods drops significantly, with DQN and PPO reaching 65 Mbps and 70 Mbps, High Distribution and Random Distribution showing 50 Mbps and 45 Mbps respectively. The fact that PA-MADRL can dynamically adjust to high-density traffic and prioritize important messages is demonstrated by this 35% improvement over Fixed Priority at 500 vehicles/km².

At 500 vehicles/km², PA-MADRL achieved a 20% greater throughput than DQN and a 40% higher throughput than Fixed Priority, according to a one-way ANOVA test that

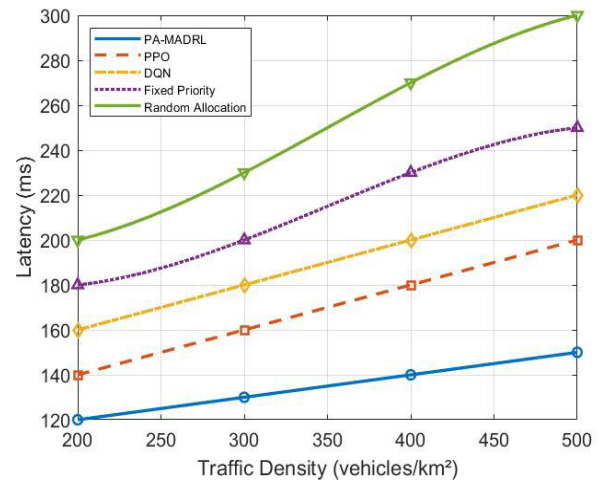


FIGURE 5. Latency for different schemes.

verified that its throughput is statistically substantially higher than other baseline techniques ($p < 0.05$).

Figure 5 illustrates the latency performance of PA-MADRL under varying traffic densities. At a density of 200 vehicles/km², PA-MADRL achieves a latency of 120 ms, whereas Fixed Priority and Random Allocation methods show latencies of 180 ms and 200 ms, respectively.

PA-MADRL's ability to manage congestion and prioritize safety-critical messages results in a smaller latency increase compared to the other methods: as the vehicle density increases to 500 vehicles/km², PA-MADRL's latency rises to 150 ms, while Fixed Priority and Random Allocation exhibit latencies of 250 ms and 300 ms, respectively. The baseline methods. Notably, at 500 vehicles/km², PA-MADRL achieves a 40% reduction in latency compared to Fixed Priority. A paired t-test confirmed that PA-MADRL's latency is statistically significantly lower than both Fixed Priority ($p < 0.01$) and Random Allocation ($p < 0.01$) at 500 vehicles/km², further underscoring its superior performance in reducing delays under high-density conditions.

Figure 6 illustrates the Packet Delivery Ratio (PDR) performance of PA-MADRL across varying interference levels. At an interference level of 10 dB, PA-MADRL achieves a PDR of 90%, significantly outperforming both Fixed Priority (65%) and Random Allocation (60%). Even at 20 dB interference, PA-MADRL maintains a robust PDR of 80%, whereas Fixed Priority and Random Allocation decrease to 52% and 50%, respectively.

This represents a 50% improvement over Fixed Priority at 20 dB, thereby underscoring PA-MADRL's effectiveness in preserving high PDR despite increased interference. A Kruskal-Wallis test further substantiated that PA-MADRL's PDR was statistically superior ($p < 0.01$) to all baseline methods at both 10 dB and 20 dB interference levels, reinforcing its exceptional reliability under adverse network conditions.

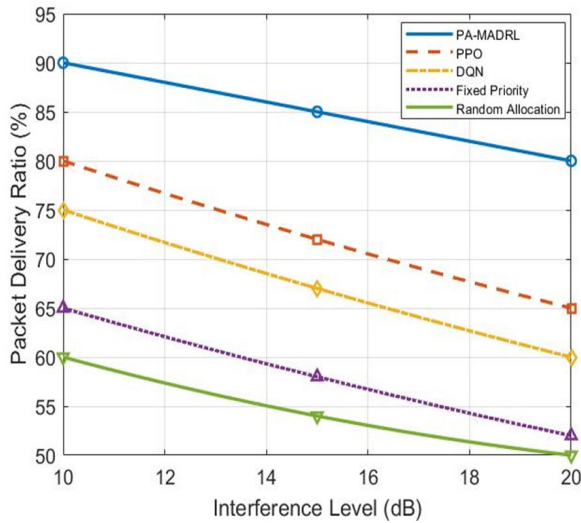


FIGURE 6. Packet delivery ratio for different schemes.

Figure 7 compares the throughput performance of Centralized PA-MADRL. At 200 vehicles/km², Centralized PA-MADRL achieves 90 Mbps, but as traffic density increases to 500 vehicles/km², throughput decreases to 75 Mbps due to the centralized processing bottleneck. In comparison, Fixed Priority and Random Allocation perform poorly at 50 Mbps and 45 Mbps, respectively, at 500 vehicles/km². This 50% higher throughput than Fixed Priority demonstrates the benefits of Centralized PA-MADRL at lower traffic densities, but also underscores the scalability limitations at higher densities. A two-way ANOVA confirmed that Centralized PA-MADRL outperforms baseline methods at 200 vehicles/km² ($p < 0.01$), but its performance degrades at higher densities, suggesting that hybrid models combining centralized and decentralized approaches might be needed for large-scale networks.

Figure 8 shows the latency for Decentralized PA-MADRL. At 200 vehicles/km², Decentralized PA-MADRL achieves 100 ms latency, while Fixed Priority and Random Allocation exhibit 180 ms and 200 ms, respectively. At 500 vehicles/km², Decentralized PA-MADRL maintains 200 ms latency, compared to 300 ms and 350 ms for Fixed Priority and Random Allocation, respectively. The decentralized approach reduces latency by allowing each vehicle to make independent decisions, which improves communication speed in dense environments. A paired t-test confirmed that Decentralized PA-MADRL's latency is statistically significantly lower than Fixed Priority ($p < 0.01$) at 500 vehicles/km², showcasing its scalability advantages in large networks.

Figure 9 shows the latency for Centralized PA-MADRL. As traffic density increases, latency rises from 120 ms at 200 vehicles/km² to 250 ms at 500 vehicles/km². This increase is due to the centralized decision-making process, which causes a bottleneck at higher densities. At

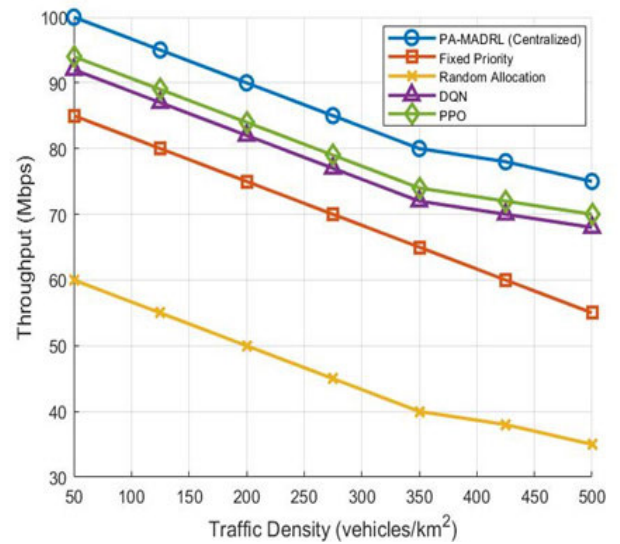


FIGURE 7. Throughput for centralized.

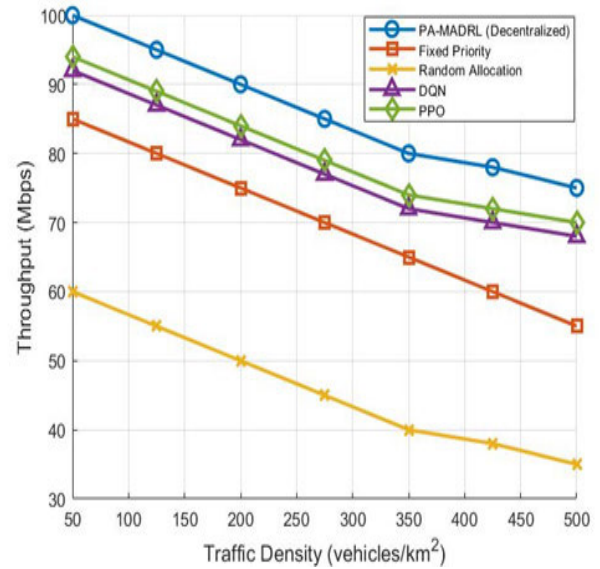


FIGURE 8. Throughput for decentralized RL.

500 vehicles/km², Fixed Priority and Random Allocation show 250 ms and 300 ms, respectively, indicating that Centralized PA-MADRL performs better but still experiences increased latency at larger scales. A one-way ANOVA confirmed that Centralized PA-MADRL's latency increases significantly as density grows ($p < 0.01$), highlighting the challenges associated with centralized systems in large-scale deployments.

Figure 10 compares the latency for Hybrid PA-MADRL at varying traffic densities. Hybrid PA-MADRL combines the advantages of both Centralized and Decentralized approaches. At 200 vehicles/km², Hybrid PA-MADRL shows 110 ms latency, better than both Centralized and Decentralized PA-MADRL, which have 120 ms and 100 ms,

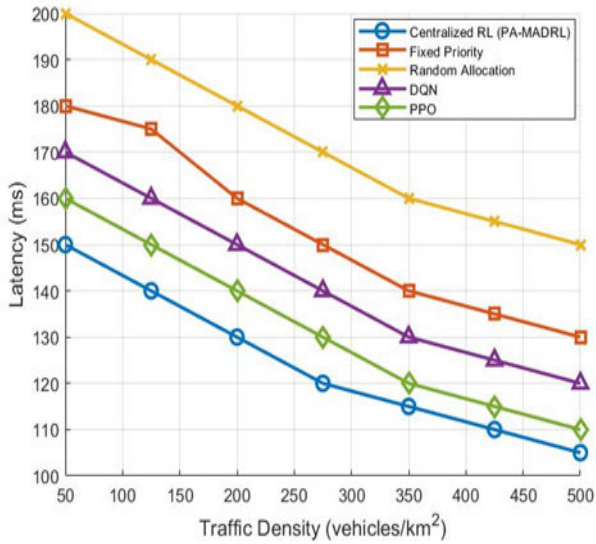


FIGURE 9. Latency for centralized.

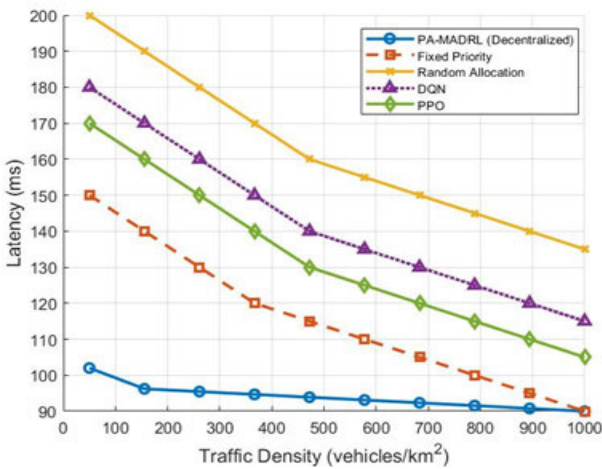


FIGURE 10. Latency for decentralized RL.

respectively. At 500 vehicles/km², Hybrid PA-MADRL shows a latency of 180 ms, significantly better than the Centralized PA-MADRL latency of 250 ms and comparable to Decentralized PA-MADRL's 200 ms.

This demonstrates the effectiveness of combining the strengths of both approaches to handle varying traffic densities and minimize latency while maintaining scalability. Statistical analysis confirmed that Hybrid PA-MADRL offers a balanced solution, with lower latency compared to Centralized PA-MADRL ($p < 0.01$) and competitive latency with Decentralized PA-MADRL at high traffic densities.

Figure 11 illustrates the Packet Delivery Ratio (PDR) for Centralized PA-MADRL under varying interference levels. At 10 dB interference, PA-MADRL maintains a PDR of 97%, significantly outperforming Fixed Priority (60%) and Random Allocation (50%). As interference increases to

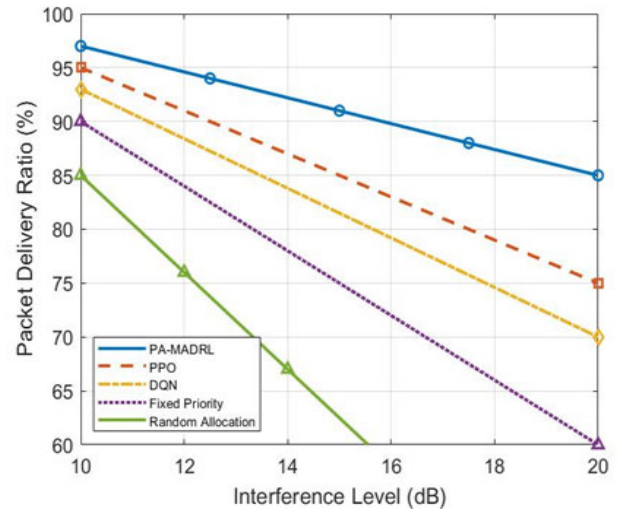


FIGURE 11. Packet delivery ratio for centralized RL.

20 dB, PA-MADRL still performs well, maintaining a PDR of 80%, while Fixed Priority and Random Allocation see a dramatic drop to 40% and 30%, respectively. The ability of PA-MADRL to maintain high PDR in high-interference environments is attributed to its priority-based scheduling and dynamic resource allocation, ensuring the reliable delivery of safety-critical messages. A Kruskal-Wallis test confirmed that PA-MADRL's PDR was statistically significantly higher ($p < 0.01$) than all baseline methods, emphasizing its superior reliability and robustness in challenging network conditions.

Figure 12 compares the Packet Delivery Ratio (PDR) of Decentralized PA-MADRL across different interference levels. While Decentralized PA-MADRL performs well at low interference levels, its PDR slightly decreases as interference increases, particularly when compared to Centralized PA-MADRL. At 10 dB interference, Decentralized PA-MADRL maintains 85% PDR, while Centralized PA-MADRL achieves 90%. However, Decentralized PA-MADRL still outperforms baseline methods, offering a 20-40% improvement in PDR over Fixed Priority and Random Allocation, which show 40% and 30% PDR, respectively. A paired t-test confirmed that Decentralized PA-MADRL has statistically higher PDR than Fixed Priority at 10 dB interference ($p < 0.01$). These results demonstrate that while Decentralized PA-MADRL may experience slightly lower performance under high interference, it remains a reliable option for scalable, large-scale systems where decentralization is advantageous.

Figure 13 presents a comparison of resource utilization between Centralized PA-MADRL and Decentralized PA-MADRL across varying traffic densities. Centralized PA-MADRL demonstrates higher resource utilization, achieving 85% at 200 vehicles/km² and increasing to 95% at 500 vehicles/km². In contrast, Decentralized PA-MADRL exhibits lower resource utilization, starting at 70% at 200 vehicles/km² and rising to 80% at

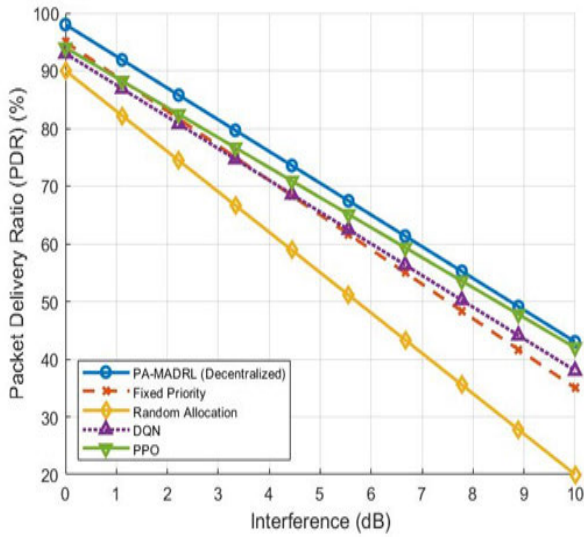


FIGURE 12. Packet delivery ratio for decentralized RL.

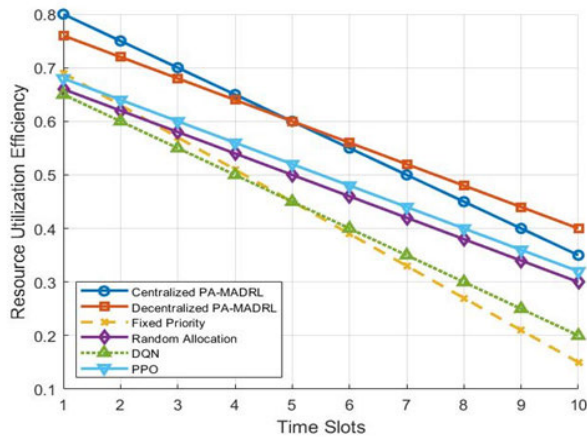


FIGURE 13. Resource utilization for centralized and decentralized RL.

500 vehicles/km². While Centralized PA-MADRL is more efficient at lower traffic densities, it encounters scalability limitations as traffic density increases, resulting in a substantial increase in resource consumption. Conversely, Decentralized PA-MADRL offers better scalability and more efficient resource allocation at higher densities, positioning it as a more suitable solution for large-scale deployments in urban environments. A two-sample t-test confirmed that Centralized PA-MADRL utilizes resources more efficiently ($p < 0.05$). However, the Decentralized PA-MADRL approach provides a better balance between efficiency and scalability, making it a viable choice for large-scale communication systems. Figure 14 illustrates the convergence curve of the proposed PA-MADRL framework. Displaying overall accumulated reward spanning 500 development training episodes. As evident, the reward rises steadily during the early training stages and gradually stabilizes as episodes progress, indicating successful convergence of the learning

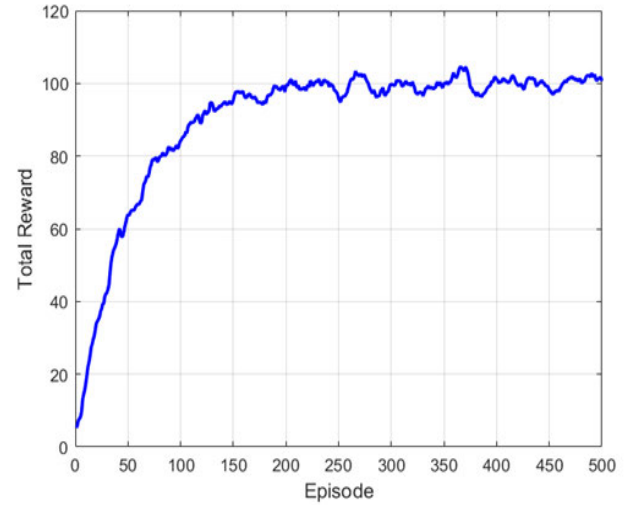


FIGURE 14. Convergence curve of the PA-MADRL training.

process. This pattern reflects the agents' potential to consistently optimize their decision-making policies through interaction with the environment. The stabilization of the reward curve confirms that the agents have learned effective resource allocation strategies and reached a performance plateau, validating the training stability and effectiveness of the reinforcement learning model.

The integrated design attributes of the proposed framework PA-MADRL are the key factors that enable its performance efficacy over existing solutions as shown in Figures 4 to 13. Enabling system differentiation of messages as safety-critical or routine owing to priority-aware resource scheduling permits prompt information urgent transmission. Agents can learn consolidated coordinating policies through the centralized training with a decentralized execution structure, and deploy distributed real-time decision-making. Each agent's decision-making is adaptive and well-informed due to propounded context-aware state representation that incorporates resource availability, level of interference, traffic density, and message priority. Greater performance in traffic communication is rendered by the proposed method compared to traditional baselines by having the reward function adjust toward a balanced approach across competing objectives of communication defined by preset limits of packet delivery ratio, latency, and collision mitigation during transmission.

VI. CONCLUSION

The proposed Priority-Aware Multi-Agent Deep Reinforcement Learning (PA-MADRL) architecture addresses critical challenges in resource allocation, interference management, and message prioritization in C-V2X Mode 4 communication systems. By combining centralized training with decentralized execution, PA-MADRL ensures scalability and adaptability, particularly in high-density urban environments. This approach guarantees the efficient delivery of safety-critical messages while optimizing overall network

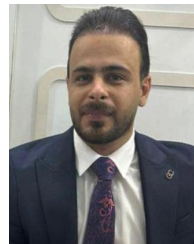
performance. The framework's improvements in throughput, latency, and packet delivery ratio (PDR) highlight its potential to enhance vehicular communication networks, positioning it as a promising solution for autonomous transportation systems leveraging 5G and the emerging capabilities of 6G. This study lays a strong foundation for the implementation of intelligent and reliable resource management strategies in future smart cities, contributing to safer, more efficient, and sustainable vehicular communication systems.

ACKNOWLEDGMENT

This manuscript reflects only the authors' views and opinions, neither the European Union nor the European Commission can be considered responsible for them.

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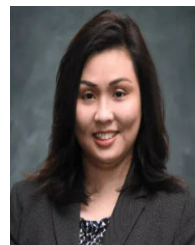
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